OpenAlex Audit of Datasets Published by CU Boulder Researchers

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Table of contents

1 CU Scholar **1**

2 Datasets authored by CU researchers in other venues **3**

3 Exploratory Analysis and Visualization **10**
3.1 Most cited datasets with a CU author 10
3.2 Repository destinations for CU-authored datasets 11
3.3 Number of dataset publications by CU author 13
3.4 Total dataset citations, by CU author 16

```
library(tidyverse)
library(openalexR)
```

```
# Load OpenAlex API key
options(openalexR.apikey = Sys.getenv("OPENALEX_API_KEY"))
```

1 CU Scholar

We start by extracting CU Scholar datasets in OpenAlex. First, we'll extract the CU Scholar's OpenAlex ids.

```
# extracts CU Scholar source information
cu_scholar_source<-oa_fetch(
  entity = "sources",
  search = "CU Scholar",
  verbose = TRUE)
```

```
# creates new column with clean IDs
cu_scholar_source<-cu_scholar_source %>%
                      mutate(clean_id=basename(id))

# extracts id column from "cu_scholar_source"
combined_ids<-cu_scholar_source$clean_id

# collapse ids into "or" string
combined_ids_string<-paste(combined_ids, collapse="|")
```

Now, we'll use the ID information to extract CU Scholar datasets:

```
# extract cu scholar datasets and assign the result to
# "cu_scholar_datasets_base"
cu_scholar_datasets_base <- oa_fetch(
  entity = "works",
  "locations.source.id" = combined_ids_string,
  type = "dataset",
  verbose = TRUE)
```

We will write this dataset to a file named *cu_scholar_datasets_base.csv*:

```
# writes "cu_scholar_datasets_base" to disk
write_csv(cu_scholar_datasets_base,
          "data-outputs/cu_scholar_datasets_base.csv")
```

`cu_scholar_datasets_base` contains 470 records/datasets, which is significantly less than the 657 datasets reported in the latest State of Open report. This might be driven by a mismatch between how objects are classified in CU Scholar and how they are classified by Open Alex (i.e. the "dataset" filter might be overly restrictive if OpenAlex interprets some CU Scholar datasets as articles instead). An alternate strategy for pulling CU Scholar datasets from OpenAlex is to extract all of the CU Scholar works indexed by OpenAlex, and then parse the "landing_page_url" field for the word "data":

```
# extract all cu scholar works
cu_scholar_works <- oa_fetch(
  entity = "works",
  "locations.source.id" = combined_ids_string,
  verbose = TRUE)
```

```
# extract cu scholar datasets by searching for "data" within
# the "landing_page_url" field, i.e. using a url parsing strategy
cu_scholar_datasets_alternate<-cu_scholar_works %>%
                                filter(str_detect(landing_page_url, "data"))
```

The `cu_scholar_datasets_alternate` data frame contains 533 records, which is substantially more than `cu_scholar_datasets_base`; it looks like many of the "Media and Climate Change" datasets were classified by OpenAlex as "articles" or "other", so they were dropped when using the "dataset" filter but picked up when parsing the landing page urls. However, the number of datasets in `cu_scholar_datasets_alternate` is still well well short of the number of datasets documented in the State of Open report. It could be useful to cross-reference the dataset(s) of CU Scholar datasets from OpenAlex with the dataset used for State of Open to try and identify which datasets are "missing", and why they are not being picked up by OpenAlex.

We'll write out `cu_scholar_datasets_alternate` to disk as "cu_scholar_datasets_alternate.csv":

```
# writes "cu_scholar_datasets_alternate" to disk
write_csv(cu_scholar_datasets_alternate,
          "data-outputs/cu_scholar_datasets_alternate.csv")
```

2 Datasets authored by CU researchers in other venues

To extract datasets published by CU researchers in any location, we'll extract CU Boulder's institutional ID in Open Alex, and then carry out a dataset search within `oa_fetch()` using this ID:

```
# extracts CU Boulder institutional information and assigns to
# "cu_boulder_metadata"
cu_boulder_metadata <-
  oa_fetch(
    entity = "institutions",
    search = "University of Colorado Boulder")

# extracts CU Boulder openalex institutional ID and assigns it to "cu_inst_id"
cu_inst_id <-basename(cu_boulder_metadata$id[1])

# extracts cu-authored datasets in any location using "cu_inst_id"
all_cu_datasets <- oa_fetch(
  entity = "works",
  "authorships.institutions.id" = cu_inst_id,
```

```
  type = "dataset",
  verbose = TRUE)
```

The resulting dataset, `all_cu_datasets`, contains 5171 datasets from 74 unique repositories. The repositories in the dataset are the following:

```
# prints unique repos in "all_cu_datasets"
unique(all_cu_datasets$source_display_name)
```

```
 [1] "Harvard Dataverse"
 [2] "Carbon Dioxide Information Analysis Center (CDIAC) Datasets"
 [3] "Worldwide Protein Data Bank"
 [4] "Water Resources Research"
 [5] "Figshare"
 [6] "PsycEXTRA Dataset"
 [7] "IUPAC Standards Online"
 [8] "Zenodo (CERN European Organization for Nuclear Research)"
 [9] "IIASA PURE (International Institute of Applied Systems Analysis)"
[10] "Recherche Data Gouv France"
[11] "University of Groningen research database (University of Groningen / Centre for Informa
[12] "NSF Seismological Facility for the Advancement of Geoscience (SAGE)"
[13] NA
[14] "Artifact Digital Object Group"
[15] "Default Digital Object Group"
[16] "XRDS Crossroads The ACM Magazine for Students"
[17] "Global Monitoring Laboratory"
[18] "ICPSR Data Holdings"
[19] "Digital USD (University of San Diego)"
[20] "Forest Service Research Data Archive"
[21] "Bristol Research (University of Bristol)"
[22] "Lasso NBDC Data Repository"
[23] "UC Santa Barbara"
[24] "CU Scholar (University of Colorado Boulder)"
[25] "DRYAD"
[26] "Publishing Network for Geoscientific and Environmental Data (PANGAEA) (Alfred Wegener I
[27] "Earth Observing Laboratory"
[28] "DIGITAL.CSIC (Spanish National Research Council (CSIC))"
[29] "DOE Oak Ridge National Laboratory (ORNL) Repository"
[30] "Borealis"
[31] "Mendeley Data"
[32] "Otago University Research Archive (University of Otago)"
```

```
[33] "National Institute of Standards and Technology (NIST)"
[34] "DOE Lawrence Berkeley National Laboratory (LBNL) Repository"
[35] "CaltechDATA"
[36] "National Oceanic and Atmospheric Administration (NOAA) National Centers for Environment
[37] "KNAW research portal (Royal Academy of Art and Sciences (KNAW))"
[38] "Institutional Repositories DataBase (IRDB)"
[39] "ScienceDB"
[40] "HEPData"
[41] "CGSPace A Repository of Agricultural Research Outputs (Consultative Group for Internati
[42] "Neotoma Paleoecological Database"
[43] "Texas Digital Library (University of Texas)"
[44] "energyo"
[45] "JPL Data"
[46] "OSTI OAI (U.S. Department of Energy Office of Scientific and Technical Information)"
[47] "HydroShare Resources"
[48] "Texas Advanced Computing Center"
[49] "USGS DOI Tool Production Environment"
[50] "Max Planck Digital Library"
[51] "National Institutes of Health, National Institute on Aging Genetics of Alzheimer's Disc
[52] "The Catalogue of Life"
[53] "Space Physics Data Facility"
[54] "European Synchrotron Radiation Facility"
[55] "CUAHSI"
[56] "Ministry of Culture and Education"
[57] "ALBA Synchrotron"
[58] "Purdue University Research Repository"
[59] "DOE Pacific Northwest National Laboratory (PNNL) Repository"
[60] "Socio-Environmental Systems Modeling"
[61] "NERC Environmental Data Service"
[62] "Institut Laue-Langevin"
[63] "UC San Diego"
[64] "ZBW Journal Data Archive"
[65] "Murdoch University Research Portal"
[66] "Stanford Digital Repository"
[67] "Small Bodies Node SBN of NASA's Planetary Data System (PDS)- Archives"
[68] "Strasbourg Astronomical Data Center"
[69] "Research Data Repository, Duke University"
[70] "Biological and Chemical Oceanography Data Management Office"
[71] "Global Biodiversity Information Facility"
[72] "Data Commons"
[73] "PolyPublie (École Polytechnique de Montréal)"
[74] "Data portal"
```

This search picks up a relatively small number (29) of CU Scholar datasets; it's not clear why, but OpenAlex may be having issues with reading the institutional ID of the depositor when it is entered into DataCite. It may also be driven by the fact that we only started explicitly tagging the institutional affiliation of the depositor in DataCite relatively recently.

We can delete the CU Scholar records in `all_cu_datasets` to generate a new dataset of CU-researcher-authored dataset publications in venues outside CU Scholar (`all_cu_datasets_sans_cuscholar`), which we'll write out to disk as "all_cu_datasets_sans_cuscholar.csv"):

```
# extracts cu-authored datasets in repositories besides cu scholar
all_cu_datasets_sans_cuscholar<-
  all_cu_datasets %>%
   filter(source_display_name!="CU Scholar (University of Colorado Boulder)")

# writes "all_cu_datasets_sans_cuscholar" to disk
write_csv(all_cu_datasets_sans_cuscholar,
          "data-outputs/all_cu_datasets_sans_cuscholar.csv")
```

We can combine the search results from the CU Scholar repository search (`cu_scholar_datasets_alternate`) with the results from the author search that excludes CU Scholar results (`all_cu_datasets_sans_cuscholar`) to generate a more comprehensive dataset that includes OpenAlex-indexed datasets from CU Scholar and other repositories that were published by CU authors (`cu_datasets_combined`):

```
# dataset that combines "cu_scholar_datasets_alternate" and
# "all_cu_datasets_sans_cuscholar"
cu_datasets_combined<-
  bind_rows(cu_scholar_datasets_alternate, all_cu_datasets_sans_cuscholar) %>%
    distinct(id, .keep_all = TRUE)
```

We'll write this dataset (which has 5590 records) to disk as "cu_datasets_combined.csv":

```
# writes "cu_datasets_combined" to disk
write_csv(cu_datasets_combined,
          "data-outputs/cu_datasets_combined.csv")
```

It's important to note that this is still likely a significant under-count of the number of datasets associated with CU researchers, since researchers do not always list institutional affiliations in the relevant metadata field when depositing datasets in external repositories (especially if there are many authors, and the CU author is not the depositing or corresponding author). We can try to build a more inclusive search net by extracting a dataset of CU-affiliated researchers (with researcher IDs), and then searching for datasets associated with these individuals. In particular, we'll use the "last_known_institutions.id" attribute to return IDs for authors whose last known publication is associated with CU Boulder:

```r
# fetch author profiles tied to cu boulder
cu_authors <- oa_fetch(
  entity = "authors",
  "last_known_institutions.id" = cu_inst_id,
  verbose = TRUE)
```

Now, we'll extract OpenAlex author IDs from `cu_authors` and assign these ids to a vector named `cu_author_ids`:

```r
# Extract authors' unique OpenAlex Author IDs
cu_author_ids <- basename(cu_authors$id)
```

Now, we'll write a function, `fetch_dataset_chunk()` to extract datasets associated with the author IDs of researchers of interest.

```r
# write processing function that takes a vector of researcher IDs and returns
# datasets associated with those ids
options(timeout = 600)
fetch_dataset_chunk <- function(id_vector) {
  Sys.sleep(1)   # pause before each request
  combined_author_string <- paste(id_vector, collapse = "|")

  oa_fetch(
    entity = "works",
    "authorships.author.id" = combined_author_string,
    type = "dataset",
    verbose = TRUE
  )
}
```

Before applying `fetch_dataset_chunk()` to `cu_author_ids`, we must chunk `cu_author_ids` into groups of 50, since OpenAlex limits batched OR filters. We'll assign the resulting list of chunked IDs (where each list element is a chunk of ~50 ids) to `author_chunks`:

```r
# chunk author ID vector into list with groups/elements of ~50 author ids
author_chunks <- split(cu_author_ids, ceiling(seq_along(cu_author_ids) / 100))
```

Finally, we'll iteratively pass the ID chunks in `author_chunks` to the `fetch_dataset_chunk`, and bind the elements of the resulting output list into a dataframe that removes any duplicated datasets:

```
# use purrr to map over the chunks using "fetch_dataset_chunk"
cu_researcher_datasets_master <- author_chunks %>%
  map(fetch_dataset_chunk) %>%
  list_rbind() %>%
  distinct(id, .keep_all = TRUE)
```

We will write `cu_researcher_datasets_master` to disk:

```
write_csv(cu_researcher_datasets_master,
          "data-outputs/cu_researcher_datasets_master.csv")
```

To create the most comprehensive dataset of published datasets associated with CU, we'll append the records in `cu_researcher_datasets_master` and those in `cu_datasets_combined`, while removing any duplicated records. We'll assign the resulting dataset to `cu_datasets_comprehensive`.

```
# creates "cu_datasets_comprehensive" by appending
# "cu_researcher_datasets_master" and "cu_datasets_combined"
cu_datasets_comprehensive<-
  bind_rows(cu_researcher_datasets_master, cu_datasets_combined) %>%
  distinct(id, .keep_all = TRUE)
```

We'll generate a new column, named "authors", that prints all of the authors associated with a dataset, separated by a semicolon (currently, the author metadata is nested within the "authorships" column, and therefore inaccessible when exporting the data from R):

```
# creates columns for author name, open alex id, and ORCID id from
# author data in "authorships" column
cu_datasets_comprehensive<-cu_datasets_comprehensive %>%
  mutate(
    authors_name=map_chr(authorships,
                         function(x){
                           x$display_name %>%
                           na.omit() %>%
                           str_c(collapse="; ")
                         }
                        ),
    authors_oa_id=map_chr(authorships,
                          function(x){
                           x$id %>%
                            na.omit() %>%
```

```
                            str_c(collapse="; ")
                          }
                        ),
    authors_orcid_id=map_chr(authorships,
                             function(x){
                              x$orcid %>%
                               na.omit() %>%
                               str_c(collapse="; ")
                             }
                        )

                    )
```

We'll also generate analogous columns that extract the CU-affiliated authors of the datasets:

```
cu_datasets_comprehensive<-cu_datasets_comprehensive %>%
  mutate(
    cu_authors_name=map_chr(authorships,
                            function(x){
                              x %>%
                                filter(basename(id) %in% cu_author_ids) %>%
                                pull(display_name) %>%
                                na.omit() %>%
                               str_c(collapse="; ")
                            }
                           ),
    cu_authors_oa_id=map_chr(authorships,
                             function(x){
                               x %>%
                                 filter(basename(id) %in% cu_author_ids) %>%
                                 pull(id) %>%
                                 na.omit() %>%
                                 str_c(collapse="; ")
                             }
                           ),
    cu_authors_orcid_id=map_chr(authorships,
                                function(x){
                                x %>%
                                 filter(basename(id) %in% cu_author_ids) %>%
                                   pull(orcid) %>%
                                   na.omit() %>%
                                   str_c(collapse="; ")
```

```
                            }
                        )
                    )
```

We'll now write out `cu_datasets_comprehensive` to disk:

```
# writes "cu_datasets_comprehensive" to disk
write_csv(cu_datasets_comprehensive,
          "data-outputs/cu_datasets_comprehensive.csv")
```

3 Exploratory Analysis and Visualization

We'll use `cu_datasets_comprehensive` as the basis for some basic exploratory analysis and visualization, since it's the most comprehensive dataset we've generated.

3.1 Most cited datasets with a CU author

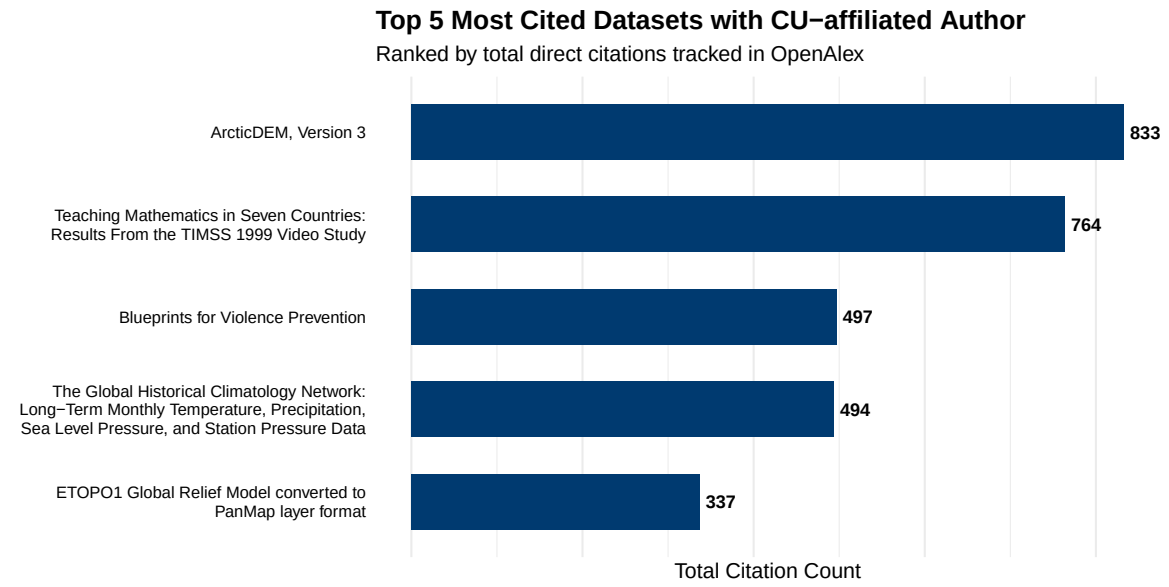
First, we'll plot the titles of the five most cited datasets with a CU author:

```
# generates dataset of top five papers
top_papers_dataset<-
  cu_datasets_comprehensive %>%
              select(title, cited_by_count) %>%
              filter(!is.na(title)) %>%
              slice_max(cited_by_count, n=5) %>%
              mutate(wrapped_title=str_wrap(title, width=45))


# plots top 5 papers
top_papers_plot<-
  ggplot(data=top_papers_dataset,
         aes(x=cited_by_count, y=fct_reorder(wrapped_title, cited_by_count)))+
  geom_col(fill="#003A70", width=0.6)+
  geom_text(aes(label=cited_by_count), hjust=-0.2, fontface="bold", size=4)+
  theme_minimal(base_size=13)+
  labs(
    title="Top 5 Most Cited Datasets with CU-affiliated Author",
    subtitle="Ranked by total direct citations tracked in OpenAlex",
    x="Total Citation Count",
```

```
    y=""
  )+
  theme(panel.grid.major.y=element_blank(),
        plot.title=element_text(face="bold", size=16),
        axis.text.x = element_blank(),
        axis.text.y=element_text(lineheight=0.9, color="black")
  )


# prints "top_papers_plot"
top_papers_plot
```



3.2 Repository destinations for CU-authored datasets

Now, we'll examine the number of CU-authored datasets by repository/publication venue:

```
# generates dataset that indicates number of datasets published in
# each repo/venue
dataset_count_by_repo<-
  cu_datasets_comprehensive %>%
  group_by(source_display_name) %>%
  count() %>%
  arrange(desc(n)) %>%
  rename(repo=source_display_name) %>%
```

```
  ungroup()

# prints "dataset_count_by_repo"
dataset_count_by_repo
```

```
# A tibble: 186 x 2
   repo                                                                  n
   <chr>                                                             <int>
 1 DOE Pacific Northwest National Laboratory (PNNL) Repository        4663
 2 OSTI OAI (U.S. Department of Energy Office of Scientific and Technical~  2198
 3 Figshare                                                           1950
 4 The Cambridge Structural Database                                  1269
 5 Zenodo (CERN European Organization for Nuclear Research)           1065
 6 PsycEXTRA Dataset                                                   753
 7 CU Scholar (University of Colorado Boulder)                         532
 8 Harvard Dataverse                                                   301
 9 Worldwide Protein Data Bank                                         275
10 EBAS                                                                229
# i 176 more rows
```

We'll write out the dataset to disk as "dataset_count_by_repo.csv":

```
write_csv(dataset_count_by_repo,
          "data-outputs/dataset_count_by_repo.csv")
```

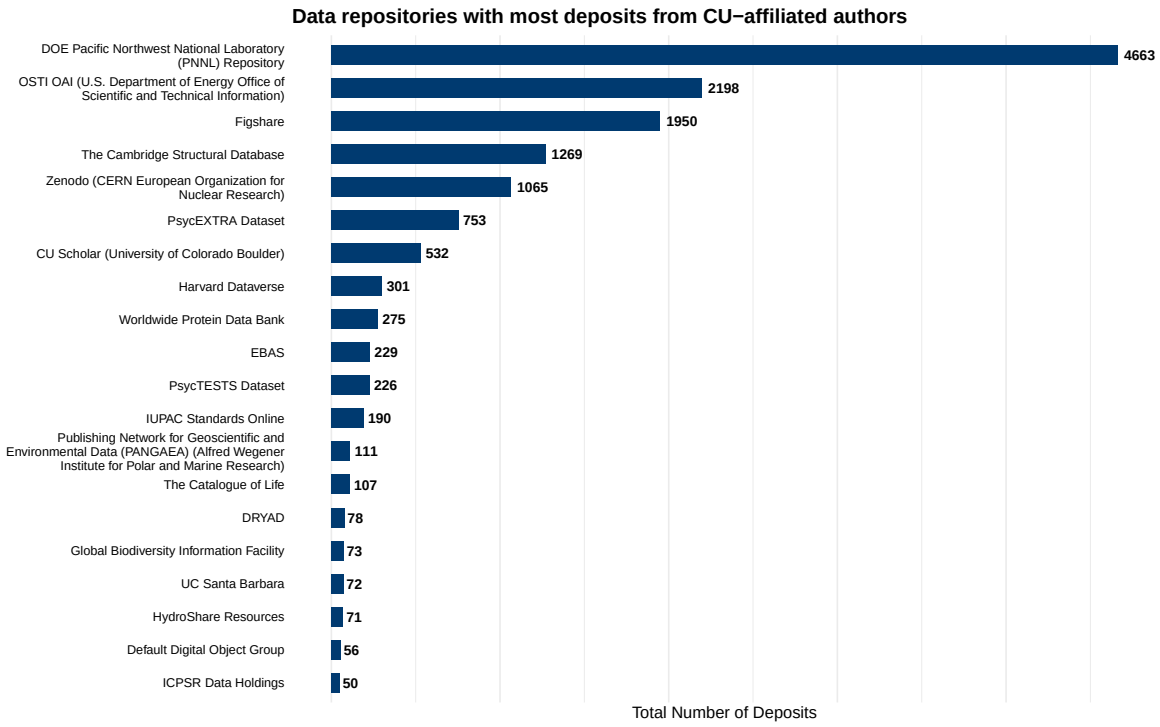
We'll also create a chart that displays the 20 most commonly used repositories:

```
# generates dataset of top ten repos
top_repos_dataset<-
  dataset_count_by_repo %>%
              filter(!is.na(repo)) %>%
              slice_max(n, n=20) %>%
              mutate(wrapped_title=str_wrap(repo, width=45))

# generates plot based on "top_repos_dataset"
top_repos_plot<-
  ggplot(data=top_repos_dataset, aes(x=n, y=fct_reorder(wrapped_title, n)))+
  geom_col(fill="#003A70", width=0.6)+
  geom_text(aes(label=n), hjust=-0.2, fontface="bold", size=4)+
  theme_minimal(base_size=13)+
  labs(
```

```
    title="Data repositories with most deposits from CU-affiliated authors",
    x="Total Number of Deposits",
    y=""
  )+
  theme(panel.grid.major.y=element_blank(),
        plot.title=element_text(face="bold", size=16),
        axis.text.x = element_blank(),
        axis.text.y=element_text(lineheight=0.9, color="black")
  )

# prints "top_repos_plot"
top_repos_plot
```



3.3 Number of dataset publications by CU author

Next, we will generate a dataset that indicates the total number of dataset publications associated with each CU author, assigned to `cu_researcher_dataset_publication_number`:

```r
# generates dataset that indicates number of dataset publications for
# each CU author
cu_researcher_dataset_publication_number<-
  cu_datasets_comprehensive %>%
separate_longer_delim(cols=c(cu_authors_oa_id,
                             cu_authors_name,
                             id),
                      delim="; ") %>%
   mutate(cu_authors_oa_id = trimws(cu_authors_oa_id)) %>%
  distinct() %>%
  group_by(cu_authors_oa_id)  %>%
  summarise(
    cu_authors_name=first(cu_authors_name),
    dataset_count=n(),
    .groups="drop") %>%
  filter(cu_authors_oa_id!="") %>%
  arrange(desc(dataset_count))

# prints "cu_researcher_dataset_publication_number"
cu_researcher_dataset_publication_number
```

```
# A tibble: 1,810 x 3
   cu_authors_oa_id                 cu_authors_name        dataset_count
   <chr>                            <chr>                          <int>
 1 https://openalex.org/A5100598901 Xuefei Sun                      1999
 2 https://openalex.org/A5067222746 Michael D. SanClements          1696
 3 https://openalex.org/A5044465853 Quazi Z. Rasool                  938
 4 https://openalex.org/A5124932560 Irina Petropavlovskikh           220
 5 https://openalex.org/A5032356449 Brian T. Kress                   188
 6 https://openalex.org/A5050152527 Seth R. Marder                   174
 7 https://openalex.org/A5085421098 Garrett Rue                      125
 8 https://openalex.org/A5076609099 Stephen H. Barlow                109
 9 https://openalex.org/A5004045735 Thomas M. Marchitto              107
10 https://openalex.org/A5051620437 John T. Andrews                  104
# i 1,800 more rows
```

We will write out the full dataset to disk:

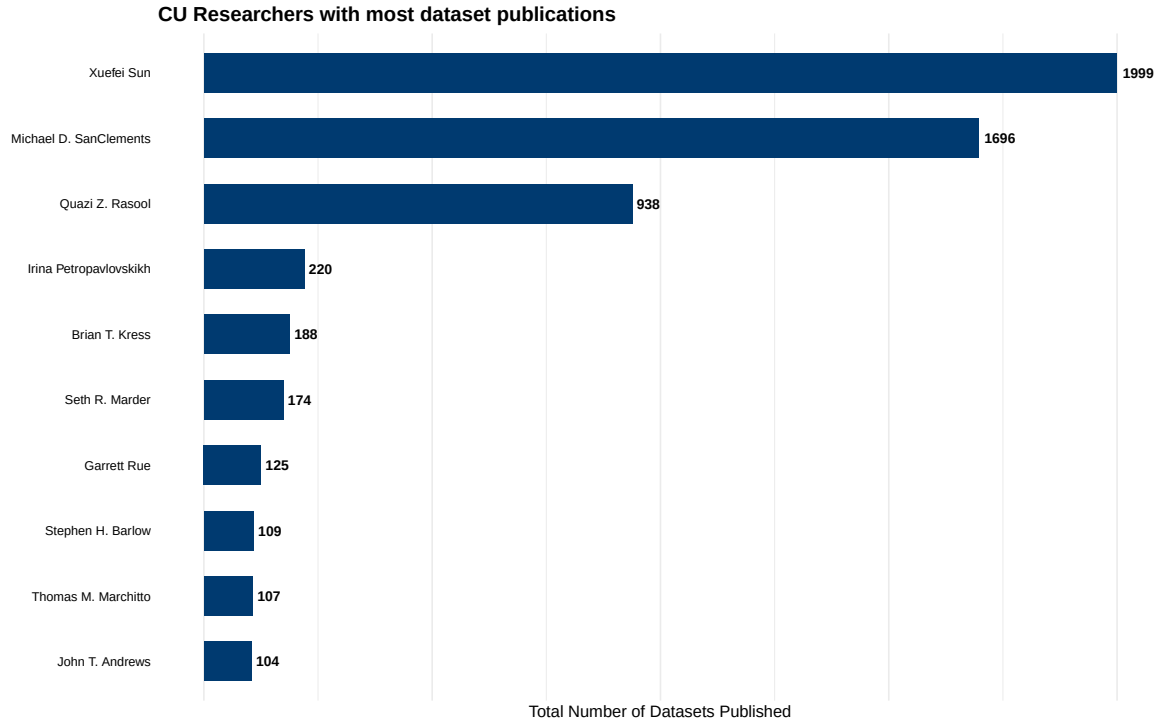
```r
# writes "cu_researcher_dataset_publication_number" to disk
write_csv(cu_researcher_dataset_publication_number,
          "data-outputs/cu_researcher_dataset_publication_number.csv")
```

Below, we plot the ten 10 CU researchers with the highest number of total dataset publications in any venue:

```
# generates dataset of top ten dataset publsihers
top_cu_dataset_publishers<-
  cu_researcher_dataset_publication_number %>%
              slice_max(dataset_count, n=10) %>%
              mutate(wrapped_title=str_wrap(cu_authors_name, width=45))

# generates plot based on "top_cu_dataset_publishers"
cu_researcher_dataset_publication_number_plot<-
  ggplot(data=top_cu_dataset_publishers,
         aes(x=dataset_count, y=fct_reorder(wrapped_title, dataset_count)))+
  geom_col(fill="#003A70", width=0.6)+
  geom_text(aes(label=dataset_count), hjust=-0.2, fontface="bold", size=4)+
  theme_minimal(base_size=13)+
  labs(
    title="CU Researchers with most dataset publications",
    x="Total Number of Datasets Published",
    y=""
  )+
  theme(panel.grid.major.y=element_blank(),
        plot.title=element_text(face="bold", size=16),
        axis.text.x = element_blank(),
        axis.text.y=element_text(lineheight=0.9, color="black")
  )

# prints "cu_researcher_dataset_publication_number_plot"
cu_researcher_dataset_publication_number_plot
```



3.4 Total dataset citations, by CU author

Finally, we will generate a dataset of the total number of dataset citations for each CU researcher:

```
# generates dataset that indicates CU authors and their total dataset citations
cu_researcher_dataset_citations<-
  cu_datasets_comprehensive %>%
  separate_longer_delim(cols=c(cu_authors_oa_id,
                               cu_authors_name,
                               id),
                        delim="; ") %>%
  mutate(cu_authors_oa_id = trimws(cu_authors_oa_id)) %>%
  distinct() %>%
  group_by(cu_authors_oa_id)  %>%
  summarise(
    cu_authors_name=first(cu_authors_name),
    number_of_dataset_citations=sum(cited_by_count),
    .groups="drop") %>%
  filter(cu_authors_oa_id!="") %>%
  arrange(desc(number_of_dataset_citations))
```

We'll write `cu_researcher_dataset_citations` to disk:

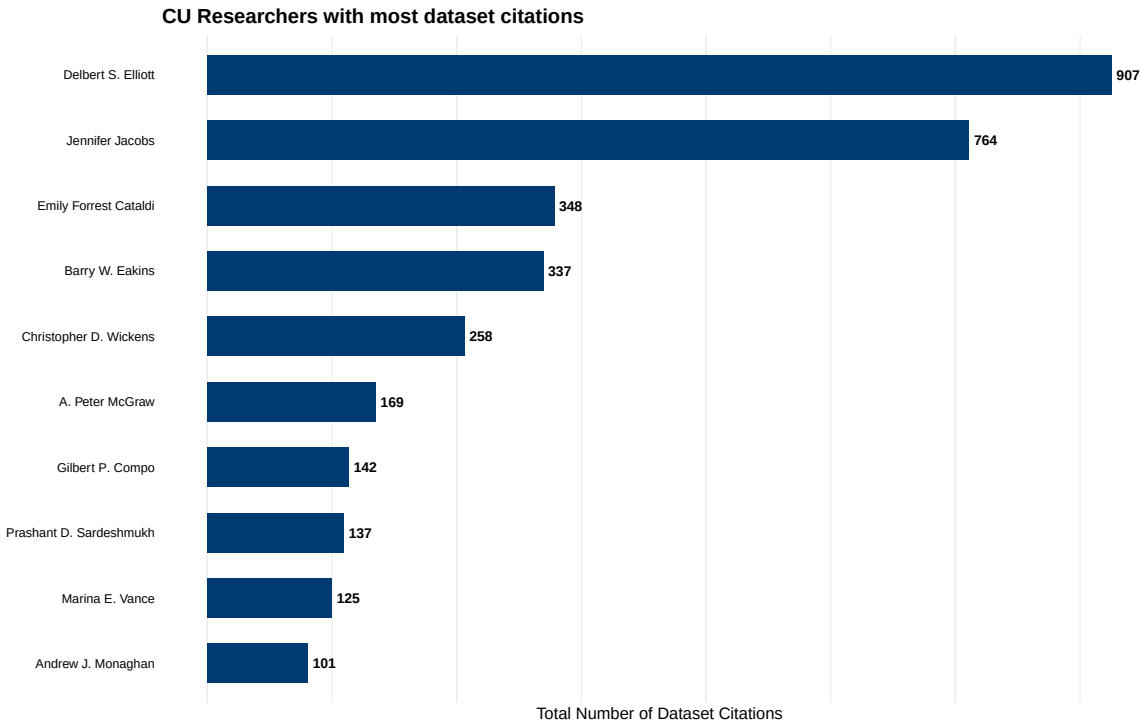
```
# writes "cu_researcher_dataset_citations" to disk
write_csv(cu_researcher_dataset_citations,
          "data-outputs/cu_researcher_dataset_citations.csv")
```

Below, we create a plot that shows CU researchers with the most dataset citations:

```
# generates dataset of top ten repos
cu_authors_with_most_data_citations<-
  cu_researcher_dataset_citations %>%
              slice_max(number_of_dataset_citations, n=10) %>%
              mutate(wrapped_title=str_wrap(cu_authors_name, width=45))

# generates plot based on "top_repos_dataset"
cu_researcher_dataset_publication_number_plot<-
  ggplot(data=cu_authors_with_most_data_citations,
         aes(x=number_of_dataset_citations,
             y=fct_reorder(wrapped_title, number_of_dataset_citations)))+
  geom_col(fill="#003A70", width=0.6)+
  geom_text(aes(label=number_of_dataset_citations),
            hjust=-0.2,
            fontface="bold",
            size=4)+
  theme_minimal(base_size=13)+
  labs(
    title="CU Researchers with most dataset citations",
    x="Total Number of Dataset Citations",
    y=""
  )+
  theme(panel.grid.major.y=element_blank(),
        plot.title=element_text(face="bold", size=16),
        axis.text.x = element_blank(),
        axis.text.y=element_text(lineheight=0.9, color="black")
  )

# prints "top_repos_plot"
cu_researcher_dataset_publication_number_plot
```


CU Researchers with most dataset citations
Delbert S. Elliott
907
Jennifer Jacobs
764
Emily Forrest Cataldi
348
Barry W. Eakins
337
Christopher D. Wickens
258
A. Peter McGraw
169
Gilbert P. Compo
142
Prashant D. Sardeshmukh
137
Marina E. Vance
125
Andrew J. Monaghan
101
Total Number of Dataset Citations